

Eenheid 1.9

Note Title

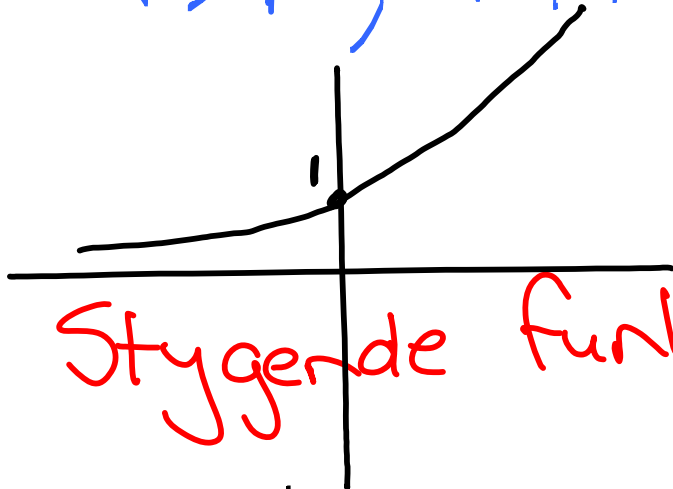
2016/02/12

Logaritmische functie

Stewart bl 59-62
Studiegids bl 14

Bekou $y = a^x$

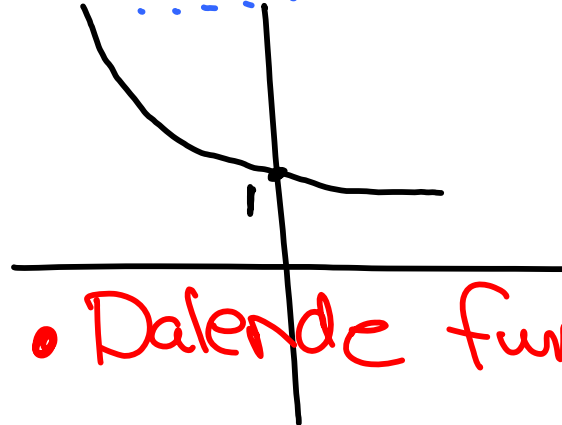
- $a > 1, a \neq 1$



- Stygende functie

$0 < a < 1$

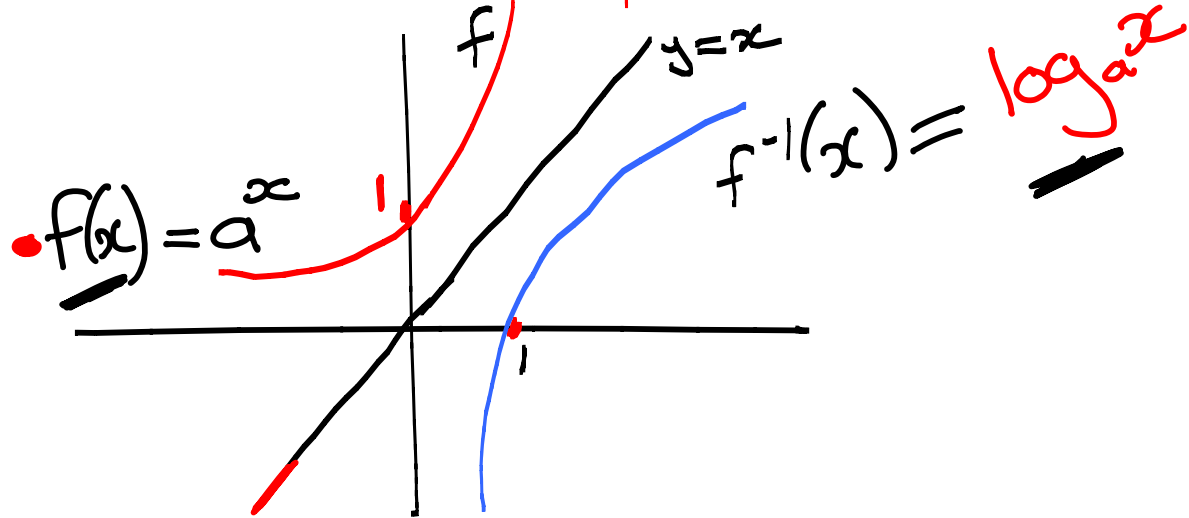
Bv $(\frac{1}{2})^x = 2^{-x}$



- Dalende functie

*Toepassing van horizontale lyn toets:

- f stygend of dalend
- f een - tot - een
- f^{-1} bestaan



Definisie:
 $x = a^y \Leftrightarrow y = \log_a x$

$$y = a^x \Leftrightarrow x = a^y$$

$$\Leftrightarrow \log_a x = \log_a a^y$$

$$\Leftrightarrow \log_a x = y \log_a a = y \cdot 1 = y$$

$$\Leftrightarrow y = \log_a x$$

Bepaal x as $\log_2 8 = x$

Def: $\log_2 8 = x$

$$\Rightarrow 2^x = 8 = 2^3$$

$$\Rightarrow x = 3$$

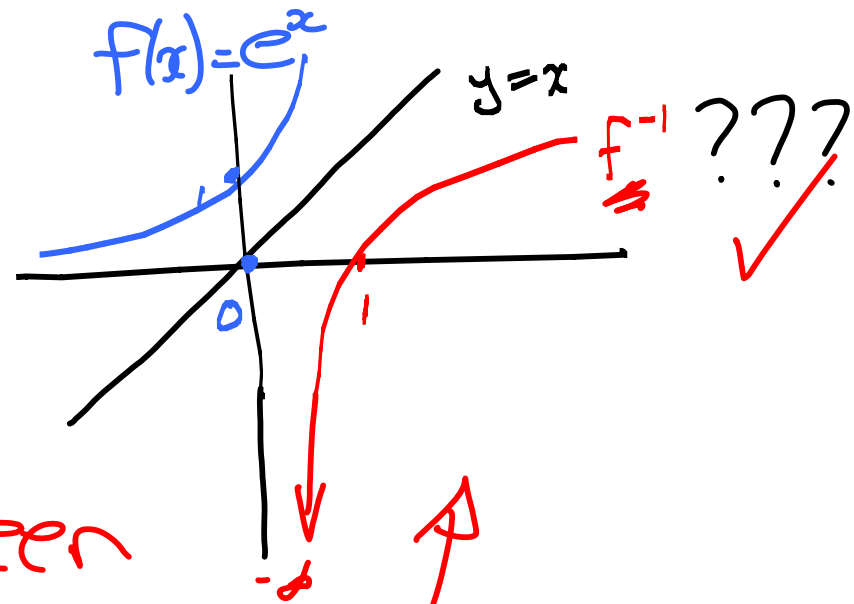
$\log_a x$: $\log_2 8 = x \log_2 2$

$$\Rightarrow \log_2 8 = x \log_2 2$$

$$\Rightarrow 8 = 2^x$$

$$\Rightarrow x = 3$$

* Natuurlijke logaritmische functie bl 60



- $y = e^x$
- $f: \mathbb{R} \rightarrow (0, \infty)$

✓ f is stijgend
✓ f is een-tot-een

Dan $f^{-1}(x)$ bestaat

$$f^{-1}: (0, \infty) \rightarrow \mathbb{R}$$

$$f^{-1}(x) = \log_e x = \ln x$$

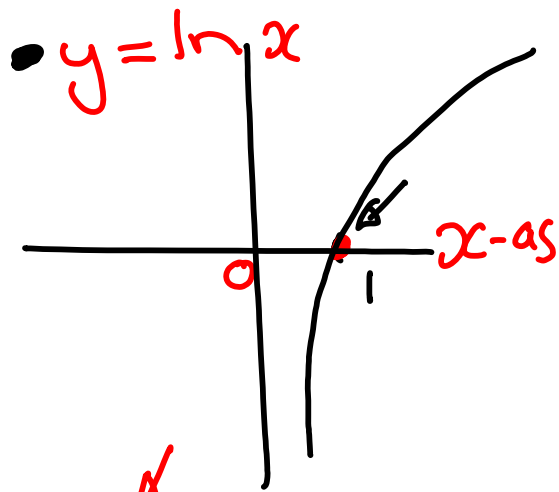
• Opmerking: $\ln e = 1$

Definition:

$$y = \ln x \Leftrightarrow e^y = x$$

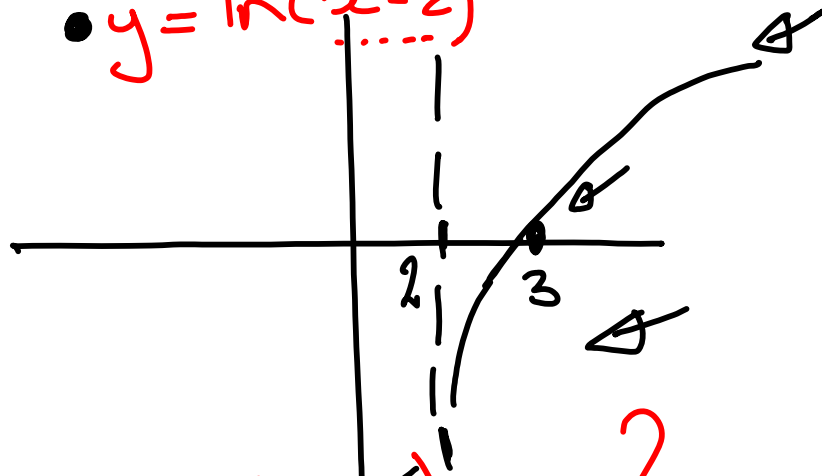
$$\begin{aligned} y = e^x &\Leftrightarrow e^y = x \\ &\Leftrightarrow \ln e^y = \ln x \\ &\Leftrightarrow y \ln e = \ln x \Leftrightarrow y = \ln x \end{aligned}$$

* Sketch: $y = -\ln(x-2)$



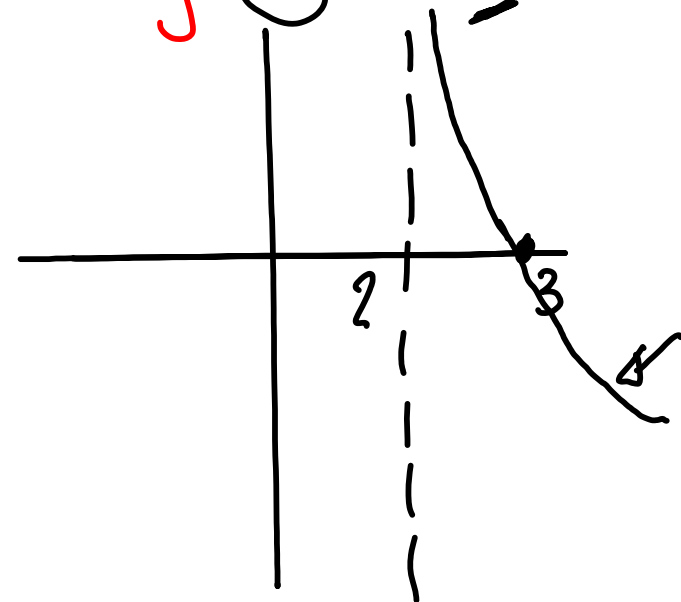
• $\ln \frac{1}{e} = 0$

• $y = \ln(x-2)$



• $\ln(x-2) = 0$?
 $\therefore x = 3$

[Onthou $y = \ln(x) \Rightarrow x > 0$]
• $y = -\ln(x-2)$



* Eienskappe van ln-funksie

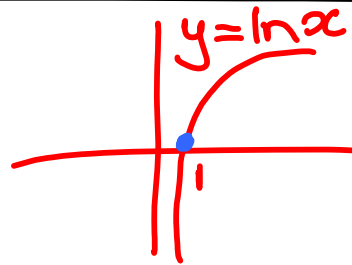
$$\textcircled{1} \ln(ab) = \ln a + \ln b \quad a, b > 0$$

$$\textcircled{2} \ln\left(\frac{a}{b}\right) = \ln a - \ln b \quad a > 0, b > 0$$

$$\textcircled{3} \ln(a^b) = b \ln a \quad a > 0$$

NB

- $\ln 1 = 0$
- $y = \ln x$ x > 0



$$D_f = (0, \infty) \checkmark$$
$$W_f = \mathbb{R} \checkmark$$

Omreken as:

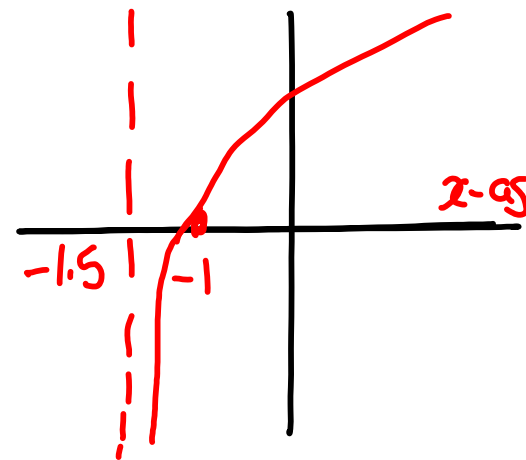
$$3 + 2x = 0$$

$$2x = -3$$

$$x = -\frac{3}{2}$$

$$= -1\frac{1}{2}$$

$$g(x) = \ln(3+2x)$$
$$\therefore 3+2x = 1$$
$$2x = -2$$
$$x = -1$$



$$\bullet g(x) = \ln(3+2x)$$

$$D_g = \left(-\frac{3}{2}, \infty\right)$$

$$W_g = \mathbb{R}$$

• kan-sellastie eien skappe / vergelykings P61

$$f(x) = e^x$$

$$D_f = \mathbb{R}, W_f = (0, \infty)$$

$$f^{-1}(x) = \ln x$$

$$D_{f^{-1}} = (0, \infty), W_{f^{-1}} = \mathbb{R}$$

$$\begin{aligned} 1) (f^{-1} \circ f)(x) &= f^{-1}(f(x)) \\ &= f^{-1}(e^x) = \ln e^x = x \ln e = x \end{aligned}$$

$x \in D_f = \mathbb{R}$

$$\ln e^x = x \text{ vir } x \in \mathbb{R}$$

$$\begin{aligned} 2) (f \circ f^{-1})(x) &= f(f^{-1}(x)) \text{ en } x \in D_{f^{-1}} = (0, \infty) \\ &= f(\ln x) = e^{\ln x} = e^y = x \end{aligned}$$

$$D_{f^{-1}} = (0, \infty)$$

$$e^{\ln x} = x \text{ vir } x > 0 \text{ (of } x \in (0, \infty))$$

$$\text{Def: } y = \ln x \Rightarrow e^y = x$$

Los op (vir x) Gelykheid $\ominus$

of $\rightarrow$ kansellasie
eienskappe
 $\rightarrow$ Definisie

$$\textcircled{1} \quad \boxed{e^{5x}} \ominus \boxed{4}$$

kansellasie: $\ln e^{5x} = \ln 4$

$$\Rightarrow 5x \ln e = \ln 4$$

$$\Rightarrow 5x = \ln 4$$

$$x = \frac{\ln 4}{5}$$

$$\begin{aligned} \text{Def: } e^{5x} &= 4 \\ \Rightarrow 5x &= \ln 4 \\ \Rightarrow x &= \frac{\ln 4}{5} \end{aligned}$$

kansel:

$$\textcircled{2} \quad e^{5x} = 4 e^{2x}$$

$$\Rightarrow \frac{e^{5x}}{e^{2x}} = 4$$

$$\Rightarrow \ln e^{3x} = \ln 4$$

$$\Rightarrow 3x = \ln 4 \Rightarrow x = \frac{\ln 4}{3}$$

$$\text{of } \ln e^{5x} = \ln (4 e^{2x})$$

$$\Rightarrow 5x \ln e = \ln 4 + \ln e^{2x}$$

$$\Rightarrow 5x = \ln 4 + 2x$$

$$\Rightarrow 3x = \ln 4$$

$$x = \frac{\ln 4}{3}$$

$$\textcircled{3} \quad \ln 2x = 8$$

Kansellasio:

$$\begin{aligned} \textcircled{\ln} 2x &= 8 \\ e^{\cancel{\ln}} 2x &= e^8 \\ \Rightarrow 2x &= e^8 \\ \Rightarrow x &= \frac{e^8}{2} \end{aligned}$$

Def. $\ln 2x = 8$

$$\Rightarrow e^8 = 2x$$

$$\Rightarrow x = \frac{e^8}{2}$$

$$\textcircled{4} \quad \ln 3x = \boxed{8} + 2 \ln x$$

Plan:

$$\Rightarrow \ln 3x - 2 \ln x = 8$$

$$\Rightarrow \ln 3x - \ln x^2 = 8$$

$$\Rightarrow \ln \left(\frac{3x}{x^2} \right) = 8$$

(kansella)

$$\Rightarrow \textcircled{e^{\ln}} \left(\frac{3}{x} \right) = e^8$$

$$\Rightarrow \frac{3}{x} = e^8 \Rightarrow \textcircled{x} = \frac{3}{e^8}$$

$$\begin{aligned} 2 \ln x &= \ln x^2 \\ \bullet \ln 3x - \ln x^2 &= \ln \left(\frac{3x}{x^2} \right) \end{aligned}$$

$$\textcircled{5} \quad e^x = 2e^2$$

(Plan)

$$\Rightarrow \frac{e^x}{e^2} = 2 \quad \text{getalle}$$

$$\Rightarrow e^{x-2} = 2$$

(Kanzel)

$$\Rightarrow \ln e^{x-2} = (\ln) 2 \Rightarrow x-2 = \ln 2$$

$$\Rightarrow x = 2 + \ln 2$$

$$\textcircled{6} \quad \ln x^2 - 6 = 0 \Rightarrow 2 \ln x = 6$$

$$\Rightarrow \ln x = \frac{6}{2} = 3$$

$$\Rightarrow e^{\ln x} = e^3 \Rightarrow x = e^3$$

$$\textcircled{7} \quad \ln 2x = 4 + 3 \ln x$$

$$\Rightarrow \ln 2x - \textcircled{3} \ln x = 4$$

$$\Rightarrow \ln 2x - \ln(x^3) = 4$$

$$\Rightarrow \ln \left(\frac{2x}{x^3} \right) = 4$$

$$\Rightarrow \textcircled{e} \ln \left(\frac{2}{x^2} \right) = e^4$$

$$\Rightarrow \frac{2}{x^2} = e^4$$

$$\Rightarrow x = \sqrt[+]{\frac{2}{e^4}} \Rightarrow x = \frac{+\sqrt{2}}{e^2}$$

maar $x > 0$

$$\boxed{\textcircled{e}^{\ln x} = x \quad \text{vir } \underline{x > 0}}$$

$$\ln a - \ln b = \ln \frac{a}{b}$$

kansellatie
eigenschap

8) As $f(x) = 1 + e^{-2x}$

• bepaal $f^{-1}(x) = y$ en • skets

$$y = f(x) \text{ en } y = f^{-1}(x)$$

$$y = 1 + e^{-2x} \Rightarrow x = 1 + \underline{e^{-2y}}$$

$$\Rightarrow \underline{\ln}(e^{-2y}) = \underline{\ln}(x - 1)$$

$$\Rightarrow -2y = \ln(x - 1)$$

$$\Rightarrow y = -\frac{\ln(x - 1)}{2}$$

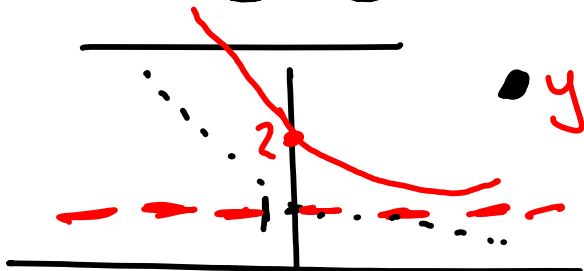
$$f(x) = 1 + e^{-2x}$$

$$f^{-1}(x) = -\frac{\ln(x - 1)}{2} \\ = -\frac{1}{2} \ln(x - 1)$$

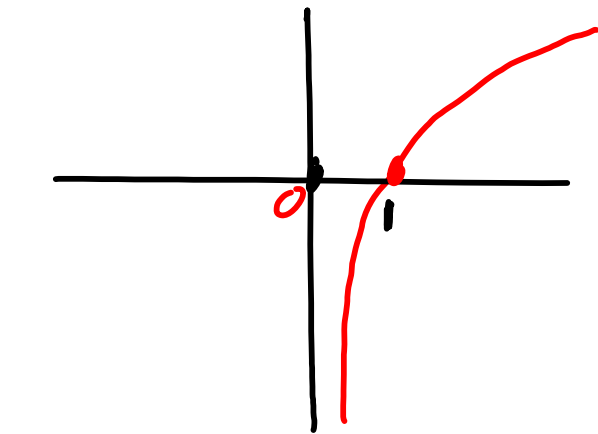
Sketch: $f(x) = 1 + e^{-2x}$ ($y = e^{-2x}$ skuif 1 eenheid op)

$y = 1 + e^{-2x}$
 $y = e^{-2x}$

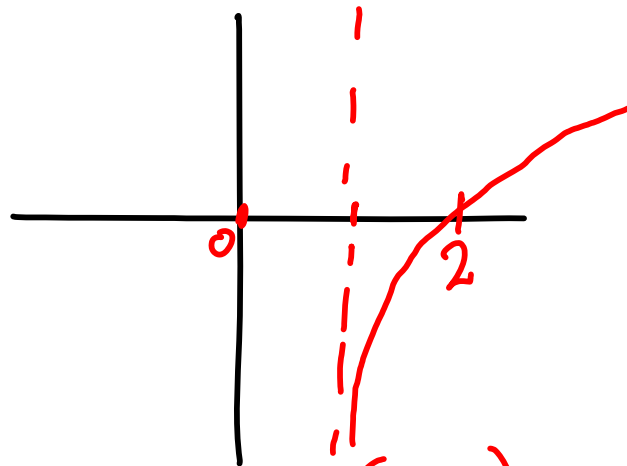
$D_f = \mathbb{R}$
 $W_f = (1, \infty)$



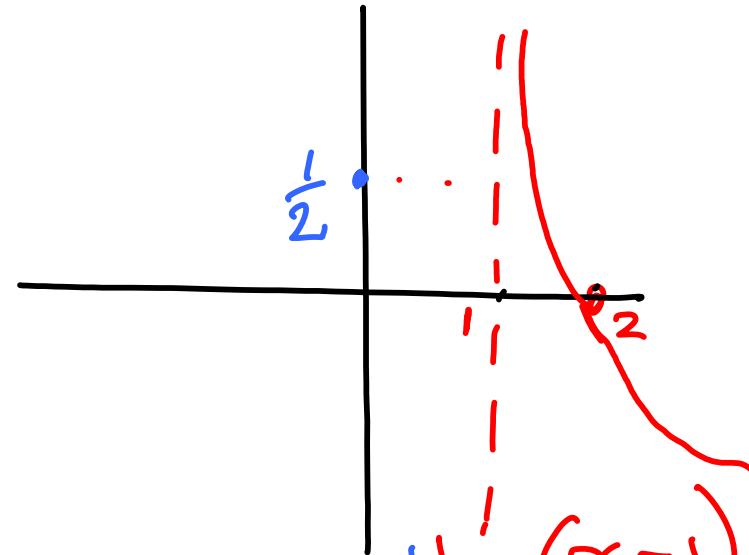
Sketch: $y = f^{-1}(x) = -\frac{\ln(x-1)}{2}$



$y = \ln x$



$y = \ln(x-1)$



$y = -\frac{1}{2} \ln(x-1)$

$W_f = D_{f^{-1}} = (1, \infty)$

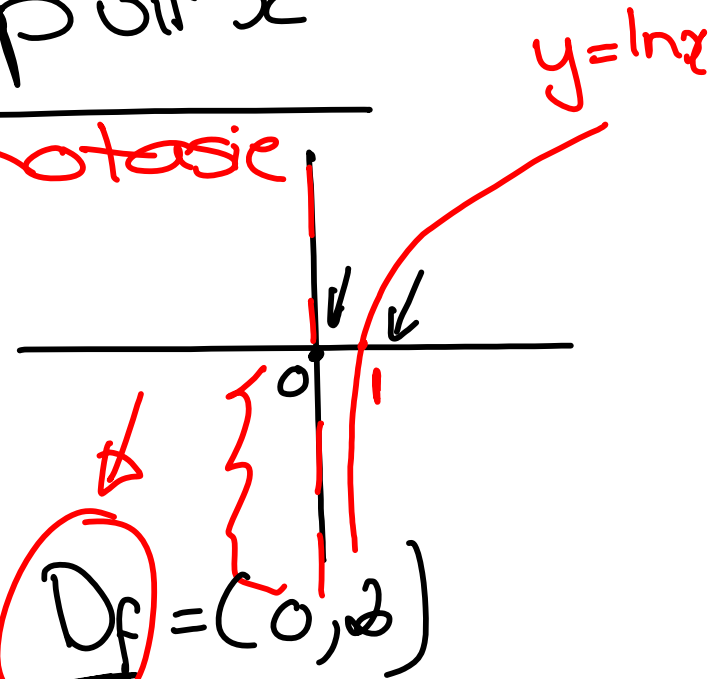
$D_f = W_{f^{-1}} = \mathbb{R}$

N3 Ongelykhede: Los op vir x

• antwoord: Intervallnotasie

① $y = \ln x < 0$
(onder x-as)

$$y = \ln x$$
$$x > 0$$



$\Rightarrow 0 < x < 1$

$\Rightarrow x \in (0, 1)$

of $e^{\ln x} < e^{0=1}$

$\Rightarrow 0 < x < 1$ Maar $D_f = (0, \infty)$

② $e^x > 5$

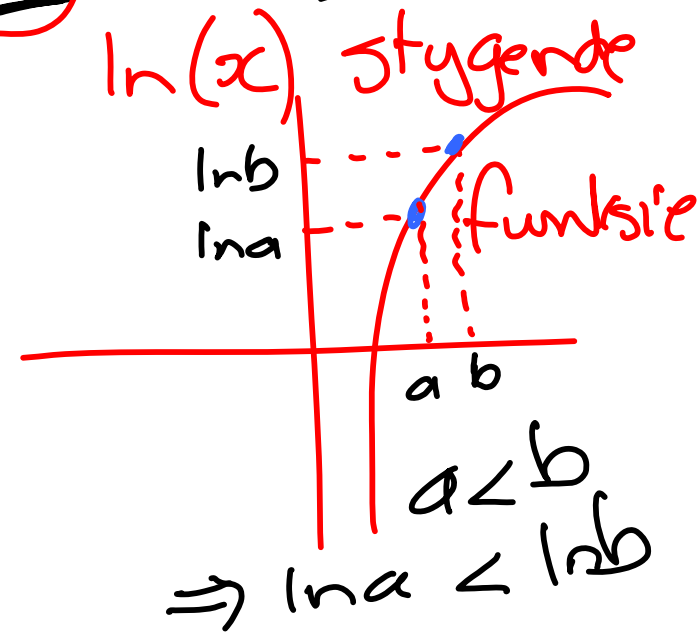
$\Rightarrow \ln e^x > \ln 5$

($\ln$ is 'n stygende funksie)

$\Rightarrow x > \ln 5$

$\Rightarrow x \in (\ln 5, \infty)$

$$y = e^x$$
$$x \in \mathbb{R}$$



$$\textcircled{3} \quad 1 < e^{3x-1} < 2$$

$$\boxed{y = e^{3x-1} \\ x \in \mathbb{R}}$$

$$D_f = \mathbb{R}$$

$$\Rightarrow \ln 1 < \ln e^{3x-1} < \ln 2$$

(ln is i stigende funktion)

$$\Rightarrow 0 < 3\underline{x} - 1 < \ln 2$$

$$\Rightarrow 0 + 1 < 3x - 1 + 1 < 1 + \ln 2$$

$$\Rightarrow \frac{1}{3} < \frac{3x}{3} < \frac{1 + \ln 2}{3}$$

$$\Rightarrow x \in \left(\frac{1}{3}, \frac{1 + \ln 2}{3} \right)$$

$$* \textcircled{4} \quad 1 - 2 \ln x \leq 3$$

$$\Rightarrow \textcircled{-2} \ln x < 3 - 1 = 2$$

$$\Rightarrow e^{\ln x} > e^{-1}$$

(e is a strictly increasing function)

$$\Rightarrow x > e^{-1} = \frac{1}{e}$$

$$\Rightarrow x \in \left(\frac{1}{e}, \infty \right)$$

$$y = \ln x$$

$$\Rightarrow x > 0$$

$$* D_f = (0, \infty) \\ = \{x \mid x > 0\}$$

⑤ $\ln x < 3$ $y = \ln x$
 $\Rightarrow x > 0$ $D_f = (0, \infty)$

$\Rightarrow e^{\ln x} < e^3$
(e is 'n stygende funksie)

$\Rightarrow x < e^3$ MAAR $x > 0$

$\Rightarrow 0 < x < e^3 \Rightarrow x \in (0, e^3)$

⑥ $f(x) = \ln(e^x - 3)$
Bepaal $\bullet D_f$ $\bullet f'(x)$ en $\bullet D_{f^{-1}}$

$\bullet D_f :$

$e^x - 3 > 0$

$\Rightarrow e^x > 3$

$y = e^x$
 $x \in \mathbb{R}$

$$\Rightarrow \ln e^x > \ln 3$$

(ln is stygende funksje)

$$\Rightarrow x > \ln 3 \Rightarrow x \in (\ln 3, \infty)$$

$$D_f = (\ln 3, \infty)$$

$$\bullet f(x) = \ln(e^x - 3)$$

$$? \bullet f^{-1}(x) = ? \quad y = ? \quad \underline{\underline{D_{f^{-1}} = ?}}$$

$$y = \ln(e^x - 3)$$

$$\Rightarrow x = \ln(e^y - 3)$$

$$\Rightarrow e^x = e^{\ln(e^y - 3)}$$

$$\Rightarrow e^x = e^y - 3$$

$$e^x > -3$$

$$\Rightarrow e^y = e^x + 3$$

$$\Rightarrow \ln e^y = \ln(e^x + 3)$$

$$\Rightarrow y = \ln(e^x + 3)$$

$$D_{f^{-1}}: e^x + 3 > 0 \text{ vis } x \in ?$$

$$\text{Dän } D_{f^{-1}} = \mathbb{R}$$

$$W_{f^{-1}} = (3, \infty)$$

$$y = e^x > 0 \text{ altyd!}$$

($\ln$ altyd ≥ 0 x -as)

